

Teaching Statement

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Teaching is central to my academic identity. It is where abstract ideas move from intangible concepts to understanding and application. As students move from curiosity to intuition, from intuition to principled inference, and from inference to implementation and clear communication, they are learning what it means to do statistics. The recurring question in my classes is simple: what generated these data, and what assumptions make our model a credible explanation of them? Across teaching assistantships at University College London and Purdue University, graduate-course support at Temple University, a guest lecture at the Wharton School, and research mentoring at Yale, my aim is to build students' technical competence together with their judgment about how to question, how to find answers, and how to communicate results responsibly.

Two philosophies that shape my teaching

My teaching reflects two complementary views about how to learn from data. From Professor Nicholas Christakis at Yale, I learned to begin with the underlying system and its data-generating process, its mechanisms, constraints, and context, so that data are read as traces of a real-world process and not as abstract columns. Paired with consequential questions and designs that make those questions answerable, empirical evidence becomes interpretable insight about the structure and dynamics of the system under study and, when warranted, about broader regularities that extend beyond a single dataset. From Professor Edoardo Airoldi at Temple University, formerly at Harvard University, I learned to make the logic of inference explicit by stating assumptions clearly, using rigorous statistical and causal reasoning, and treating probabilistic models as arguments that must be stress-tested with diagnostics, sensitivity analyses, and precise statements of uncertainty.

In practice I teach with a simple routine. First I begin with a question. Next I build intuition, articulate assumptions, and choose methods that match them. Finally I interpret results with equal attention to what worked, what failed, and why. This structure helps students see that methodology is not an end in itself. It is a disciplined way of turning a substantive question into an answer that others can scrutinize and trust.

From phenomenon to formalism

In my teaching assistant roles I have worked with both small and large groups, adjusting my methods to the size of the room and to student background. In large classes, teamwork and presentations matter for sustaining engagement. In those settings I pair each abstract object with a concrete phenomenon, then build the theory using language that makes the phenomenon precise.

In UCL's *Probability and Statistics II*, for example, I introduced distributions by contrasting patterns students can see. Heights, measured with noise, concentrate around a typical value, which motivates the normal model. Social-network degree counts and rare extremes such as wildfire sizes produce many small values and a few very large ones, which motivates heavy-tailed behavior. Only after students had articulated why these shapes differ did we formalize the mathematics, deriving expectations and variances from first principles and making the underlying independence and approximation assumptions explicit.

I apply the same questions-first, assumptions-explicit, intuition-to-formalism logic across every appointment: at University College London in *Probability and Statistics II* (2nd year), *Optimization Algorithms in Operational Research* (2nd year), *Statistical Inference* (3rd year), and *Forecasting* (3rd year); at Purdue University in *Machine Learning I*; and at Temple University, assisting Professor Airoidi with graduate *Causal Inference*. In November 2025 I gave a guest lecture in *STAT 4750, Sample Survey Methods*, at the Wharton School, presenting the design and analysis of our randomized village-network study to an audience of undergraduate and doctoral students. Teaching a live study to students who are learning survey design was a useful test of whether I can make a real design decision legible, including the parts that did not go smoothly.

Three requirements, and how I meet them

I have found three prerequisites for successful teaching: effective communication of complex material during lectures and the discussion that follows, active student engagement, and continuous practice through assignments and independent work. Whenever I enter a teaching scenario I focus on meeting these three, while staying flexible enough to adjust for different student groups.

Communication. To break down complex material, I link theory to practice and use real-world examples to anchor abstraction. I have also learned that relevance alone does not produce engagement. When I teach a module on probability and statistics, I hold a balance between intuition, real examples, mathematical derivation, and practical implementation drawn from research and from industry. I draw examples from different domains so that students with different backgrounds can each find an entry point.

Engagement. Students are often reluctant to speak if they feel pressure to already know the answer. I work to create a relaxed atmosphere by encouraging questions and by saying plainly that a student who asks a question has performed a service for the whole class. The aim is to build critical thinking while still covering the material rigorously.

Practice. I use a mix of written assignments, team assignments, and presentations, because different forms of practice develop different skills and reach students who learn in different ways. I add labs in R and Python that mirror contemporary workflows and require reproducibility, so that computation is assessed as part of the argument and not as an afterthought.

Mentoring

My mentoring style was shaped by leading research efforts in academic and industry environments, including the Human Nature Lab at Yale and Wells Fargo. In those settings I train students to replace the phrase “it is not working” with a diagnosis: which assumption is violated, where the identifying argument breaks, which diagnostic is misspecified, or what computational choice is causing the instability. I bring the same habit into teaching, because the ability to diagnose, explain, and communicate why something fails is foundational for scientific work and for responsible practice. Concretely, I meet students on a fixed schedule, I ask them to write down the estimand before they write code, and I expect every analysis to be reproducible by someone else in the group.

Courses I am prepared to teach and to develop

I am prepared to teach across the quantitative methods sequence, at both introductory and advanced levels. This includes applied and intermediate statistical methods for education and social science research, measurement and latent variable models, multilevel and longitudinal data analysis, causal inference and evaluation methods, Bayesian statistics, statistical computing and simulation, and machine learning for research applications.

I would also like to develop a graduate elective on statistical methods for social network data, covering study design and boundary specification, measurement error in self-reported ties, descriptive and model-based analysis, network models as mediators and as outcomes, and causal inference when units interfere with one another. Course work would use real data from field studies, including the intervention data from my own research, so that students see the full path from a design decision to a defensible conclusion.

What I want students to leave with

My goal is to graduate students who can reason carefully about assumptions and uncertainty, implement methods competently, and communicate conclusions and limitations to both technical and non-technical audiences. Ultimately I see teaching as empowerment. It helps students build habits of mind that move them from data to insight, and from insight to decisions with real-world consequence.